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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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SMART UNIVERSITY LABORATORY EQUIPMENT MONITORING AND ALLOCATION SYSTEM USING C

A DESIGN ASSESSMENT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

ECA1302 - Microwave Engineering

to the award of the degree of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

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SIMATS ENGINEERING

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1. PROBLEM STATEMENT

University laboratories contain various types of equipment that must be monitored for availability, condition, usage, maintenance, and allocation. Manual management of these records can lead to errors, duplicate entries, difficulty in locating equipment, and delays in identifying shortages or maintenance requirements.

The proposed Smart University Laboratory Equipment Monitoring and Allocation System using C provides a menu-driven solution to register, update, search, sort, allocate, return, merge, and analyse laboratory equipment records. The system also stores records using file handling and generates reports to identify equipment shortages, maintenance requirements, frequently used equipment, and immediate procurement needs.

2. OBJECTIVE

The main objective of this project is to develop a menu-driven C application for efficient monitoring and management of university laboratory equipment.

The specific objectives are:

- To maintain equipment details using C structures and arrays.
- To provide searching and sorting facilities.
- To manage equipment allocation and return.
- To identify equipment shortages and maintenance requirements.
- To merge laboratory records while avoiding duplicate equipment IDs.
- To demonstrate functions, pointers, recursion, and file handling.
- To generate a consolidated laboratory equipment report.

3. REQUIREMENTS AND ENVIRONMENT USED

3.1 Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Keyboard and mouse
- Internet connection for online execution, if required

3.2 Software Requirements

- Operating System: Windows
- Programming Language: C
- Compiler: GCC
- Execution Environment: OnlineGDB / C Compiler
- Web Browser: Google Chrome or equivalent

3.3 Programming Concepts Used

The project uses the following C programming concepts:

- Variables and data types
- Operators
- Decision-making statements
- Loops
- Structures
- Arrays
- Strings
- Functions
- Pointers
- Recursion
- Searching
- Sorting
- File handling

4. DESIGN / PROPOSED SOLUTION

4.1 System Overview

The proposed system is a menu-driven laboratory equipment management application. It stores equipment information and provides different operations for monitoring and managing laboratory resources.

Each equipment record contains Equipment ID, Name, Category, Quantity, Available Units, Laboratory, Condition, Maintenance Date, and Usage Count.

4.2 Proposed System Modules

Equipment Registration Module

Allows the user to add new equipment records and prevents duplicate equipment IDs.

Equipment Update Module

Allows existing equipment availability, condition, maintenance date, and usage count to be updated.

Search Module

Searches equipment using ID, category, laboratory, or condition.

Sorting Module

Sorts equipment based on availability, usage frequency, or category.

Allocation and Return Module

Manages the allocation and return of equipment units while maintaining updated availability.

Record Merging Module

Combines equipment records from different laboratories and detects duplicate IDs.

Maintenance Analysis Module

Identifies equipment in poor condition that requires maintenance.

Status Analysis Module

Classifies equipment as Available, Low Availability, Under Maintenance, or Critical.

Report Module

Generates a consolidated report containing inventory, usage, maintenance, shortage, and procurement information.

File Handling Module

Stores and retrieves equipment records using a binary data file.

4.3 Data Structure Design

A structure named `Equipment` is used to store all equipment-related information.

The structure contains:

- `id` – Equipment identification
- `name` – Equipment name
- `category` – Equipment category
- `quantity` – Total quantity
- `available` – Available units
- `laboratory` – Laboratory name
- `condition` – Equipment condition
- `maintenanceDate` – Maintenance date
- `usageCount` – Number of times used

An array of structures is used to store multiple equipment records.

5. ALGORITHM / PSEUDOCODE / FLOWCHART

5.1 Main Algorithm

1. Start the program.
2. Load previously stored records from the file.
3. Display the main menu.
4. Accept the user's choice.

5. Perform the selected operation.
6. Update equipment records when required.
7. Generate analysis and reports when requested.
8. Save records to the file.
9. Repeat until the user selects Exit.
10. Stop.

5.2 Equipment Registration Algorithm

1. Accept equipment ID.
2. Check whether the ID already exists.
3. If duplicate, reject the record.
4. Otherwise, accept equipment details.
5. Validate quantity and availability.
6. Store the record in the structure array.
7. Display successful registration.

5.3 Search Algorithm

1. Select the search criterion.
2. Enter the required search value.
3. Compare the value with stored records.
4. Display matching equipment.
5. If searching by ID, recursion is used to locate the record.

5.4 Sorting Algorithm

1. Select the sorting criterion.
2. Compare adjacent records.
3. Swap records according to the selected criterion.
4. Continue until all records are sorted.
5. Display the sorted records.

5.5 Allocation Algorithm

1. Enter the equipment ID.
2. Search for the equipment.
3. Enter the number of units required.
4. Check equipment condition.
5. Check available units.
6. If sufficient units are available, allocate them.
7. Reduce available units and increase usage count.
8. Otherwise, display allocation denied.

5.6 Record Merging Algorithm

1. Accept records from another laboratory.
2. Check each equipment ID.
3. Compare it with existing IDs.
4. Reject duplicate IDs.

5. Add non-duplicate records to the main array.
6. Display the merging result.

5.7 Maintenance Analysis Algorithm

1. Traverse all equipment records.
2. Check the equipment condition.
3. Identify records with poor condition.
4. Display equipment requiring maintenance.

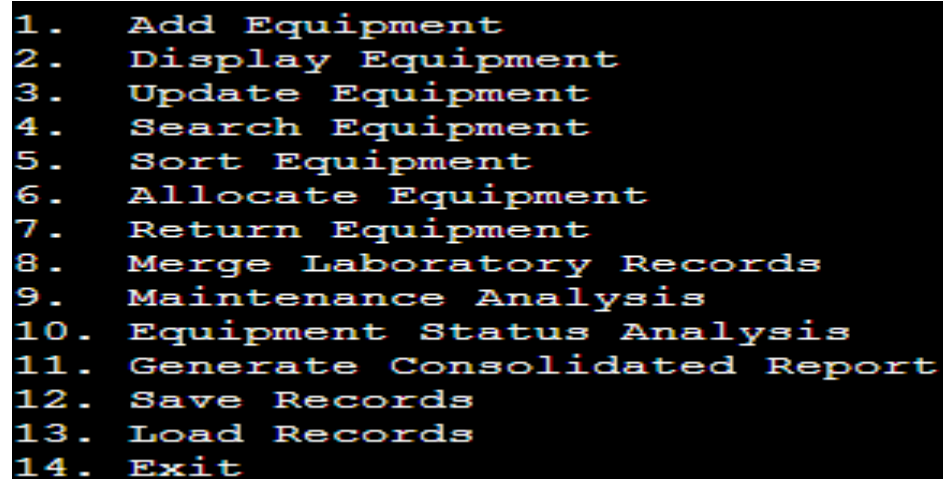
5.8 File Handling Algorithm

1. Open the data file.
2. Save the number of records and equipment data.
3. Close the file.
4. During program startup, open the file.
5. Read stored records.
6. Restore them into the equipment array.

6. IMPLEMENTATION / SOURCE CODE

The system is implemented using the C programming language. Structures are used to represent equipment records, arrays store multiple records, functions provide modularity, pointers are used for direct structure manipulation, recursion is used for ID searching, and file handling is used for permanent data storage.

The complete source code includes the following major functions:



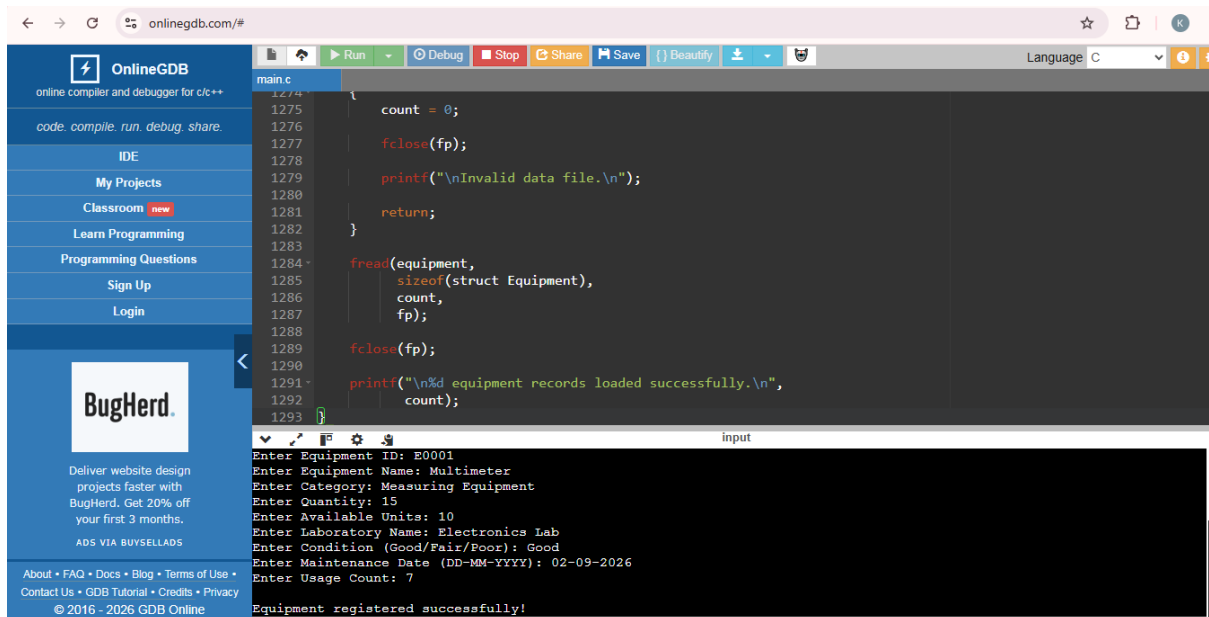
```
1.  Add Equipment
2.  Display Equipment
3.  Update Equipment
4.  Search Equipment
5.  Sort Equipment
6.  Allocate Equipment
7.  Return Equipment
8.  Merge Laboratory Records
9.  Maintenance Analysis
10. Equipment Status Analysis
11. Generate Consolidated Report
12. Save Records
13. Load Records
14. Exit
```

7. TEST CASES AND EXPECTED/ACTUAL RESULTS

Test Case	Operation	Input/Condition	Expected Result	Actual Result
TC01	Add Equipment	New Equipment ID	Record added successfully	Successful
TC02	Add Equipment	Duplicate ID	Duplicate record rejected	Successful
TC03	Search	Valid Equipment ID	Matching record displayed	Successful
TC04	Search	Valid Category	Matching equipment displayed	Successful
TC05	Search	Valid Laboratory	Laboratory equipment displayed	Successful
TC06	Sort	Sort by availability	Records sorted by available units	Successful
TC07	Sort	Sort by usage	Most-used equipment displayed first	Successful
TC08	Allocate	Sufficient units available	Equipment allocated successfully	Successful
TC09	Allocate	Insufficient units	Allocation denied	Successful
TC10	Return	Valid equipment and units	Available quantity updated	Successful
TC11	Merge	New equipment ID	Record merged successfully	Successful
TC12	Merge	Duplicate equipment ID	Duplicate record skipped	Successful
TC13	Maintenance	Poor condition	Maintenance requirement displayed	Successful
TC14	File Handling	Save records	Records stored in file	Successful
TC15	File Handling	Load records	Stored records retrieved	Successful
TC16	Report	Multiple equipment records	Consolidated report generated	Successful

8. EXECUTION SCREENSHOTS / OUTPUT

The following screenshot is included as evidence of program execution:

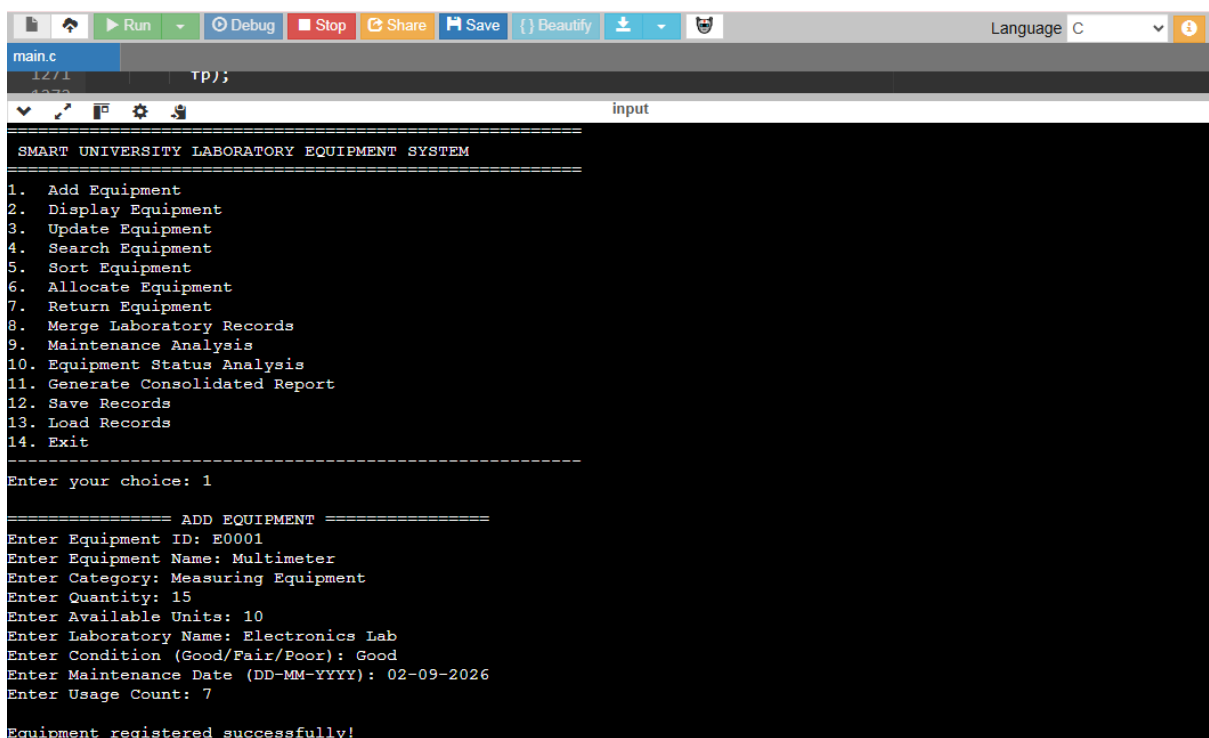


The screenshot shows the OnlineGDB IDE interface. The left sidebar contains navigation links like 'IDE', 'My Projects', 'Classroom', 'Learn Programming', 'Programming Questions', 'Sign Up', and 'Login'. The main editor displays a C program named 'main.c' with the following code:

```
1274
1275     count = 0;
1276
1277     fclose(fp);
1278
1279     printf("\nInvalid data file.\n");
1280
1281     return;
1282 }
1283
1284 fread(equipment,
1285        sizeof(struct Equipment),
1286        count,
1287        fp);
1288
1289 fclose(fp);
1290
1291 printf("\n%d equipment records loaded successfully.\n",
1292        count);
1293
```

The output window shows the program's execution results:

```
input
Enter Equipment ID: E0001
Enter Equipment Name: Multimeter
Enter Category: Measuring Equipment
Enter Quantity: 15
Enter Available Units: 10
Enter Laboratory Name: Electronics Lab
Enter Condition (Good/Fair/Poor): Good
Enter Maintenance Date (DD-MM-YYYY): 02-09-2026
Enter Usage Count: 7
Equipment registered successfully!
```



The screenshot shows the program's menu-driven interface. The output window displays the following menu:

```
===== SMART UNIVERSITY LABORATORY EQUIPMENT SYSTEM =====
1. Add Equipment
2. Display Equipment
3. Update Equipment
4. Search Equipment
5. Sort Equipment
6. Allocate Equipment
7. Return Equipment
8. Merge Laboratory Records
9. Maintenance Analysis
10. Equipment Status Analysis
11. Generate Consolidated Report
12. Save Records
13. Load Records
14. Exit
=====
Enter your choice: 1

===== ADD EQUIPMENT =====
Enter Equipment ID: E0001
Enter Equipment Name: Multimeter
Enter Category: Measuring Equipment
Enter Quantity: 15
Enter Available Units: 10
Enter Laboratory Name: Electronics Lab
Enter Condition (Good/Fair/Poor): Good
Enter Maintenance Date (DD-MM-YYYY): 02-09-2026
Enter Usage Count: 7
Equipment registered successfully!
```

Screenshot demonstrate the successful execution of the implemented functions and validate the working of the proposed system.

9. ANALYSIS AND DISCUSSION

The developed system successfully provides a centralized approach for managing laboratory equipment records. The use of structures allows different equipment attributes to be stored together, while arrays support management of multiple records.

Searching and sorting operations make it easier to locate and analyse equipment. Allocation and return functions maintain updated availability information. The merging function allows records from different laboratories to be combined while preventing duplicate IDs.

Pointers are used for direct manipulation and analysis of structure records, while recursion is demonstrated through equipment ID searching. File handling provides permanent storage and retrieval of records.

The system also identifies low-availability, critical, and maintenance-required equipment. The consolidated report supports laboratory administrators in making decisions related to equipment allocation, maintenance, replacement, and procurement.

9.1 Advantages

- Simple menu-driven interface.
- Centralized equipment management.
- Faster searching and sorting.
- Prevents duplicate equipment IDs.
- Tracks equipment usage and availability.
- Identifies maintenance requirements.
- Supports laboratory shortage analysis.
- Provides permanent data storage.
- Generates consolidated reports.

9.2 Limitations

- The system is primarily console-based.
- It does not provide a graphical user interface.
- Data is maintained locally using a file.
- Multiple users cannot access the database simultaneously.
- Authentication and user access control are not included.

9.3 Future Enhancements

- Development of a graphical user interface.
- Integration with a relational database.
- User authentication and role-based access.
- Automated maintenance reminders.
- Cloud-based laboratory equipment management.
- Barcode or QR-code based equipment tracking.

10. CONCLUSION

“The Smart University Laboratory Equipment Monitoring and Allocation System using C” successfully provides an efficient method for managing university laboratory equipment. The system supports registration, updating, searching, sorting, allocation, returning, merging, maintenance analysis, status analysis, and report generation.

The project demonstrates important C programming concepts including structures, arrays, strings, functions, pointers, recursion, searching, sorting, decision-making, loops, and file handling. The system can help laboratories improve equipment utilization, identify shortages, manage maintenance requirements, and support timely procurement decisions.

11. INDIVIDUAL CONTRIBUTION OF GROUP MEMBERS

Member 1

Equipment Management

- Add Equipment
- Display Equipment
- Update Equipment
- Save and Load Records

Member 2

Equipment Operations – Kaviyaasri M [192312614]

- **Search Equipment** – Search and find specific laboratory equipment using equipment ID or equipment name.
- **Sort Equipment** – Arrange equipment records systematically based on equipment details for easy management.
- **Allocate Equipment** – Assign available laboratory equipment to students or faculty members and store allocation details.
- **Return Equipment** – Record the return of allocated equipment and update its status as available.
- **Merge Laboratory Records** – Combine equipment records from different laboratories into a single organized record.

Member 3

Analysis and Reporting

- Maintenance Analysis
- Equipment Status Analysis
- Generate Consolidated Report
- Testing and output verification

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12. REFERENCES

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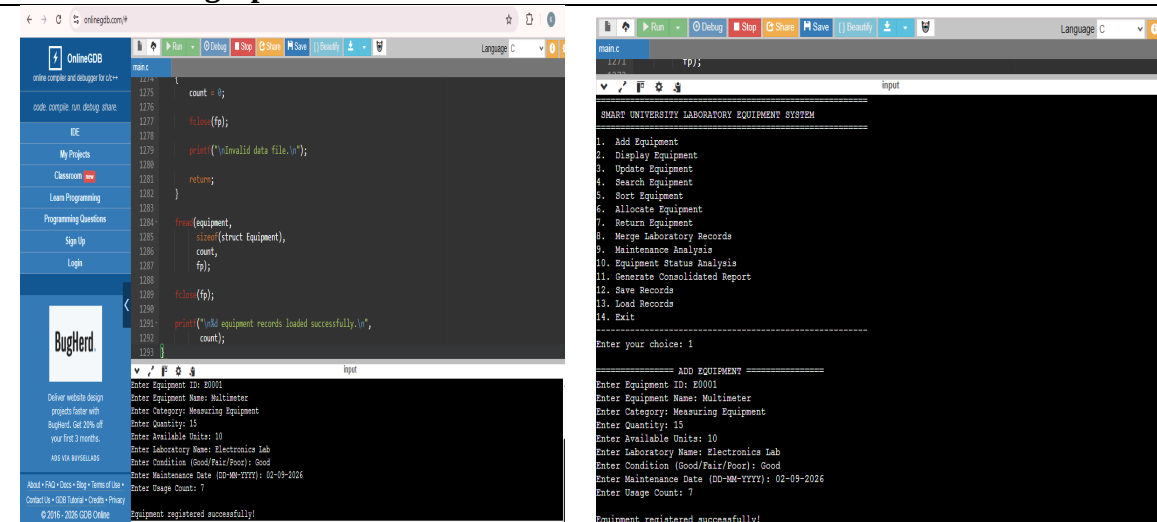
Slot: B

Course Name: CSA0211

Course Faculty: Dr Saravanan S K

Project Title: SMART UNIVERSITY LABORATORY EQUIPMENT MONITORING AND ALLOCATION SYSTEM USING C

Module Photographs:



Project Description: Module 2 - Equipment Operations

The project “**Smart University Laboratory Equipment Monitoring and Allocation System Using C**” focuses on developing an efficient system for managing, monitoring, and allocating laboratory equipment within a university environment. The system helps maintain equipment availability, manage user requests, and ensure proper utilization of laboratory resources. In **Module 2 – Equipment Allocation and Request Management**, the system is designed to handle equipment requests from students and faculty members and allocate the required equipment based on its availability. When a user submits a request, the system verifies the equipment inventory and checks whether the requested equipment is currently available for allocation. If the equipment is available, it is assigned to the respective user, and details such as **user name, user ID, equipment ID, allocation time, and expected return date** are recorded. The module also manages the return of allocated equipment and automatically updates its availability status after the equipment is returned. This module helps prevent duplicate allocation, maintains accurate equipment usage records, and ensures efficient utilization and tracking of laboratory resources. It forms an important part of the proposed system by providing organized equipment allocation and request management before the monitoring and maintenance operations are performed by the subsequent module.

Kaviyaasri M
Student Signature

Guide Signatur

